Data Exploration & Preparation - CA1
Salary Survey Analysis

Introduction

For this work we are using a salary survey where we are going to try and discern some basic information with EDA process mainly, who where gender of respondents, what's the average salary in euros for all of them, counting on the extra compensations some of the participants received, how the salary is distributed between genders and if there is any correlations with how much someone wins in terms of annual salary with their age, gender, study level and work experience being inside their study field or overall work experience outside of it.

Methods

For the methods we started by using Python and a nine step process to form our EDA. We started first by rebranding salary and compensations into a single new column called ‘Total Anual Salary’ and that column was then processed into having a single currency from the total of ten currencies that they were pooled in the data, which will be euros, that new column is called ‘‘Total Annual Salary in Euros’’

Not all incomes were kept some, those who income was classified as ‘Other’ and were removed from the list since they had values that we truly do not know from which country they are or if they mentioned the currency the total of the data was less than 1% of the total.

To keep in mind not all of the participants reported having any extra income as ‘‘Compensations’’, those who didn't and who value was missing or cero where kept as that or added as a cero to not disturbed the original salary and keep it as one to one as possible to the original answer.

We the used .boxplot to remove outliers, wich is the main culprit of the most amount of information lost from the calculation and moved ahead with the plotting of the information.

Results

Here the results of the data we processed. In general most of the salary in euros is concentrated at median of a bit over seventy one thousand euros a year(Fig1.) with differences between the sex/gender of the participants when it comes to annual salary(Fig2.)

Following the last figure(Fig3.) we can see that when it comes to annual salaries, correlations between the factors are positive but almost no discernible the more the value increases in relation to how much a participant makes over a year worth of work even when counting compensations on the equation.

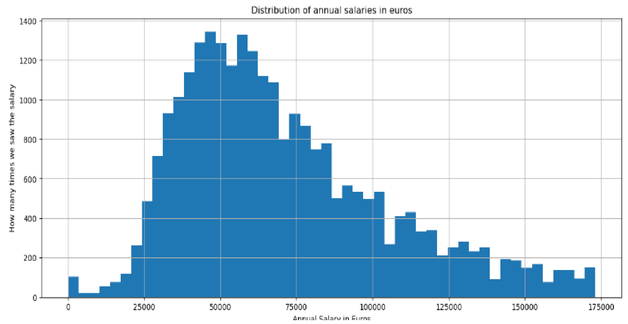


Fig. 1 Annual Salary histogram for the general population

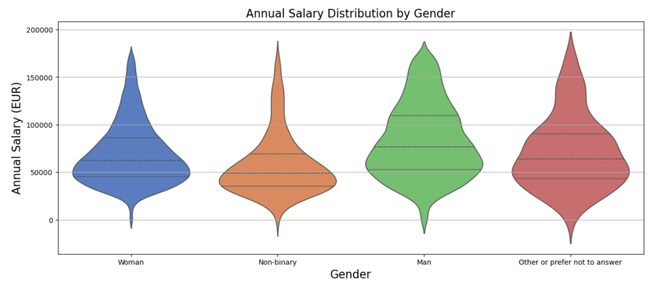


Fig. 2 Differences between genders and their annual salary

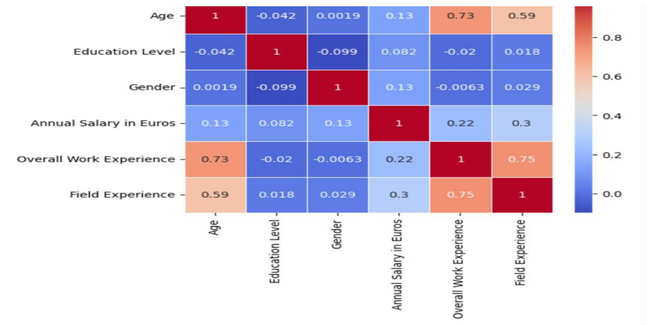


Fig. 3 Correlation heatmap

Conclusions

A couple of conclusions we can get out of the data is that:

.If we take the total of the of the data set looks like women tend to be more open to answer this survey or was put forward on channels where they were the vast majority of the participants.

.On average the total population has a annual salary counting compensations of 71.000 euros

.By gender the salary from who has the highest salary to the lowest goes for men as the highest, on average winning 82k followed by those who don't want or classify as ‘other’ at 71k a year, then women at 69k and the lowest being non-binary at 57k a year

.With the information given unfortunately we could not find any strong correlations between how much someone can make a year in relation to their age, education level, overall work experience with the highest positive relationship being ‘field experience’.

.this could tell us that at least with the this data, if someone wants to increase their yearly salary they would have a better chance of doing do if they perfect or stay longer in a single profession but the increase wont be proportional to the amount of years.

References

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